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Unit-1

Q.1 Collect Marketing Dataset

```
import pandas as pd

data = {
"product": ["Laptop", "Mobile", "Tablet"],
"Sales": [50000, 30000, 20000],
"Quantity": [5, 10, 8]
}

df = pd.DataFrame(data)
print(df)
```

=> output

	product	Sales	quantity
0	Laptop	50000	5
1	Mobile	30000	10
2	Tablet	20000	8

Q:2 output

Before

	Name	Age
0	Amit	20.0
1	Riya	21.0
2	John	NoN
3	Sara	22.0

After

	Name	Age
0	Amit	20.0
1	Riya	20.0
2	John	21.0
3	Sara	22.0

Q:7

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Q: 1 preprocess Marketing Dataset

```
import pandas as pd

data = {
    "Name": ["Amit", "Riya", "John", "Sara"],
    "Age": [20, 21, None, 22]
}

df = pd.DataFrame(data)

print("Before:")
print(df)

df["Age"] = df["Age"].fillna(df["Age"].mean())

print("\nAfter:")
print(df)
```

Q 3: output

	channel	Customers	Revenue
0	Email	100	10000
1	Instagram	150	18000
2	Email	120	12000
3	Google	200	25000

Total customers:
570

Total Revenue
65000

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Q:3 EDA on Marketing Data Set

```
Import pandas as pd

data = {
"Channel": ["Email", "Instagram", "Email", "Google"]
"Customer": [100, 120, 200],
"Revenue": [10000, 18000, 12000, 25000]
}

df = pd.DataFrame(data)

print(df).

print("\n Total customers:")
print(df["Customer"].Sum())

print("\nprint Revenue:")
print(df["Revenue"].Sum())
```

Q:4 output

A line Graph is displayed showing:

Jan → 10000
Feb → 12000
Mar → 15000
Apr → 14000
May → 18000

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Q:4 Sales Time Series plot

```
import matplotlib.pyplot as plt

Months = ["Jan", "Feb", "mar", "Apr", "May"]
Sales = [10000, 12000, 15000, 14000, 18000]

plt.plot (months, Sales, Marker = 'o')

plt.title ("monthy Sales")
plt.xable ("month")
plt.yable ("Sales")

plt.Show()
```

Q:5 output

Sales = [100, 120, 140, 160, 180]

Forecast = 160.0

Q:6 : output

predicted Sales = 200.0

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Q:5 Sales forecast using Moving.

Sales = [100, 120, 140, 160, 180]

average = sum (Sales [-3:]) / 3

print ("Sales =", Sales)
print ("forecast =" Average)

Q:6

from Sklearn.linear_model import Linear Regression

X = [[1], [2], [3], [4], [5]]
Y = [100, 120, 140, 160, 180]

Model = linear Regression ()
Model.fit (X, Y)

prediction = Model.predict ([[6]])
print ("predicted Sales =", prediction[0])

Q: output

Input

Customer	Age	Spending
A	25	500
B	30	700
C	NoN	600
D	35	800

output before preprocessing

	Customer	Age	Spending
0	A	25.0	500
1	B	30.0	700
2	C	NoN	600
3	D	35.0	800

After preprocessing

	Customer	Age	Spending
0	A	25.0	500
1	B	30.0	700
2	C	30.0	600
3	D	35.0	800

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unit 2

Q:1 preprocess customer Dataset

```
Import pandas as pd

data = {
    "Customer": ("A", "B", "C", "D"),
    "Age": [25, 30, None, 35]
    "Spending": [500, 700, 600, 800]
}

df = Pd. Data frame (data)

print ("Before preprocessing :")
print (df)

# fillmissing age with mean
df ["Age"] = df ["Age"]. fill na (df
["Age"] mean ())

print ("inAfter preprocessing :")
print (df)
```

Q:2 output :-

Input :

Age	Income	Spending	visits
20	20000	1000	2
25	30000	1500	3
30	40000	2000	4
35	50000	3000	5
40	60000	4000	6

output :

Selected features :

	Income	Spending
0	20000	1000
1	30000	1500
2	40000	2000
3	50000	3000
4	60000	4000

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Q:2 feature Selection on customer

```
import pandas as pd

data = {
"Age": [20, 25, 30, 40],
"Income": [20000, 30000, 40000, 50000, 60000],
"Spending": [1000, 1500, 12000, 3000, 4000],
"visits": [2, 3, 4, 5, 6]
}

df = pd.DataFrame(data)

print("Orignal Dataset:")
print(df)

# Select important features
features = df[["income", "Spending"]]

print("\nSelected features:")
print(features)
```

Q:3

output

input

	Customer	Recency	Frequency	Monetary
A	5	10	5000	
B	20	5	2000	
C	10	8	3500	
D	30	3	1000	

output :

RFM Analysis :

	Customer	Recency	Frequency	Monetary
0	A	5	10	5000
1	B	20	5	2000
2	C	10	8	3500
3	D	30	3	1000

Average Monetary value:

2875.0

Interpretation : customer A is the most valuable because of low recency, high frequency, and high Monetary value.

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Q:3 RFM Analysis

```
Import pandas as pd

data = {
"Customer" : ["A", "B", "C", "D"]
"Recency" : [5, 20, 10, 30]
"Frequency": [10, 5, 8, 3]
"Monetary": [5000, 2000, 3500, 1000]
}

df = Pd. Data frame (data)

print ("RFM Analysis:")
print (df)

print ("In Average Monitory value:")
print (df ["Monitory "]. mean(.)
```

Q:4 output

Income	Spending
20	20
22	25
25	22
70	75
73	80
80	78

output :

	Income	Spending	Segment
0	20	20	1
1	22	25	1
2	25	22	1
3	70	75	0
4	75	80	0
5	80	78	0

The cluster number 0 and 1 can appear in the opposite order depending on the model; the important that the customer are divided into two groups.

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Q4 Customer Segmentation using K-mean

```
Import pandas as pd
from SKlearn.cluster Import K Means

data = {
"Income": [20, 22, 25, 70, 75, 80]
"Spending": [20, 25, 22, 75, 80, 78]
}
df = PD. Data frame (data)

Model = K Means (n-clusters = 2 random-
State =
0, n-init = 10)
df = pd. Data frame (data)

Model = K Means (n-clusters = 2, random-
State = 0, n-init = 10)
df ["Segment"] = model. fit-predict (df [[
"income", "Spending"]])

print (df)
```

Q:5 output

Income	Spending	Segment
20	20	1
22	25	1
25	22	1
70	75	0
75	80	0
80	78	0

output :

a Scatter plot is displayed showing two customer segments :

Segments 1 -> low Income / low Spending
Segment 0 -> High Income / high Spending

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Q:5 visualize customer segment

```
import matplotlib.pyplot as plt

income = [20, 22, 25, 70, 75, 80]
spending = [20, 25, 22, 75, 80, 78]
segment = [1, 1, 1, 0, 0, 0]

plt.scatter(income, spending, c=segment)

plt.xlabel("Income")
plt.ylabel("Spending")
plt.title("Customer")
plt.show()
```

Q: output

Input

Number of customer = 100
Average purchase = 500
purchase per year = 4
customer lifetime = 3 years

output:
customer lifetime value = 60000
formula
CLV = Average purchase value * purchases per year * customer life time

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Q:6 Estimate customer lyfetime value (CLV)

customer =100

average purchase =500

purchase per year =4

year =3

clv : average purchase * purchase per-year *

print("customer lyfetime value :",clv)